# LLMs Encode Harmfulness and Refusal Separately A Generalization To Thinking Models

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## Abstract

Recent work demonstrates that Large Language Models (LLMs) encode harmfulness and refusal as distinct, separable concepts within their representation space. However, it remains unclear whether this representational separation persists in modern reasoning or "thinking" models, which employ extended internal chain-of-thought (CoT) processing before generating a final response. In this paper, we generalize the geometric analysis of safety representations to thinking models, investigating how the reasoning process affects the localization of harmfulness and refusal signals. Extending prior work's two-token framework into a multi-slot layout spanning the prompt, CoT, and answer positions, we find that behavioral refusal on harmful prompts remains stable (approximately 94–95%) across thinking, non-thinking, and stripped-template generation modes, while the underlying harmfulness and refusal signals become distributed across the reasoning trace rather than localized at a single boundary token. A diagnostic removing post-instruction template tokens does not increase harmful-prompt compliance, indicating that refusal in thinking models is not reducible to a single token position as in non-reasoning models. These results suggest that the harmfulness/refusal separation remains a useful lens for reasoning models, though the operational locus of the refusal decision shifts from a fixed boundary toward a distributed process across the reasoning trace.

## 1. Introduction

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